Enhancing Emotional Well-being of Middle-Aged and Older Adults through Embodied Conversational Agents: Investigating Emotional Tone and Gender Matching

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***Abstract*—With the global elderly population rapidly increasing, addressing social isolation and loneliness is essential for enhancing their quality of life. Traditional support systems often fall short due to geographical dispersion and mobility limitations, making Embodied Conversational Agents (ECAs) a promising alternative by offering companionship and emotional support through natural interactions. This study investigates how variations in ECAs' emotional tone of voice and gender-appearance congruence impact the emotional well-being and user experience of 86 participants aged 50 and above. Findings indicate that ECAs with a positive emotional tone significantly improve users' emotional states, sense of companionship, and overall satisfaction, while gender-appearance congruence enhances trustworthiness and comfort. These insights inform the design of empathetic, trustworthy, and user-centered ECAs to effectively mitigate loneliness and promote emotional well-being among the elderly.**

1. Introduction

1.1. Background

The global demographic landscape is undergoing a profound transformation marked by a notable increase in the elderly population. In the coming decades, the proportion of individuals aged 60 and older is expected to rise significantly, representing a substantial share of the global population. This demographic shift brings significant challenges, particularly in addressing the emotional and social well-being of older adults, who are disproportionately susceptible to loneliness and social isolation [1] [2] [3].

Loneliness, defined as the subjective feeling of being alone or disconnected, and social isolation, characterized by an objective lack of social interactions or relationships, are widespread among older adults and are strongly associated with a range of adverse health outcomes. Empirical studies have consistently demonstrated that these conditions are linked to heightened risks of depression, cognitive decline, cardiovascular diseases, and even premature mortality [4] [5]. These findings emphasize the critical importance of social connectedness in fostering health and well-being. Biopsychosocial mechanisms further explain how social relationships influence physiological and psychological processes, highlighting the interplay between social engagement and overall health [6] [7] [8].

Traditional support mechanisms, such as family and community networks, often fall short of addressing these challenges due to factors such as geographical separation of families, reduced mobility among the elderly, and limited access to resources. This necessitates innovative interventions to effectively mitigate loneliness and foster social engagement among this vulnerable population [9] [10] [11].

Technological advancements, particularly in artificial intelligence (AI) and human-computer interaction (HCI), have increasingly opened up innovative opportunities to address the complex social and emotional challenges experienced by older adults [12] [13] [14]. Among these innovations, Embodied Conversational Agents (ECAs) have emerged as promising tools specifically designed to deliver personalized emotional support and companionship to the elderly [15] [16]. ECAs are not merely limited to voice-based interactions; rather, they are sophisticated virtual entities that leverage multimodal communication techniques—integrating speech, facial expressions, gestures, and other non-verbal cues—to simulate human-like communication and create an engaging interaction experience [15] [17]. By harnessing these capabilities, ECAs hold significant potential to alleviate loneliness and enhance the emotional and social well-being of older adults, offering a dynamic

and adaptive solution to foster meaningful connections and improve quality of life [15] [18] [19].

1.2. Objectives

In the context of rapidly evolving human-computer interaction (HCI) paradigms, the primary aim of this study is to advance the design and application of embodied conversational agents (ECAs) tailored to support the emotional well-being of elderly users [20].

Our research is guided by three key objectives. First, we investigate the impact of an ECA's emotional tone by examining how variations in vocal delivery—specifically, comparing neutral and positively modulated speech—affect users' emotional well-being, perceptions of companionship, and overall interaction outcomes [21]. Second, we explore the role of gender and appearance matching by assessing how congruence between the agent's perceived gender through its voice and its visual representation influences trust, perceived intelligence, comfort, and satisfaction [22] [23] [24] [25]. Third, our study provides an in-depth analysis of two independent experiments, offering insights that underscore the implications for future ECA development. These insights emphasize the importance of user-centered, empathetic, and engaging designs that incorporate cultural contexts and practical design principles, thereby providing actionable guidelines for future research and development [15] [18] [19] [26].

Despite considerable prior research on ECAs and emotional speech, several critical gaps remain. For example, while concatenative synthesis is widely used, its effectiveness in conveying distinct emotional tones for emotional support has not been thoroughly evaluated. Similarly, limited studies have directly compared the impact of neutral versus happy speech on elderly users' perceptions. Moreover, the combined effects of vocal emotional tone and visual congruence on trust, perceived intelligence, and engagement require further investigation. To address these gaps, our research is driven by the following questions:

1) How does the emotional tone of an ECA's voice (neutral vs. happy) influence elderly users' emotional reactions and perceived companionship?
2) To what extent does gender–appearance congruence affect trustworthiness, perceived intelligence, and likability?
3) What are the combined effects of emotional tone and gender–appearance congruence on elderly users' overall user experience?

By systematically addressing these objectives and research questions, our study seeks to generate evidence-based insights that will inform the development of ECAs capable of mitigating loneliness and enhancing the emotional well-being and social integration of elderly users [27] [28] [29] [30].

2. Related Work

This section begins with an overview of embodied conversational agents (ECAs), focusing on their application in emotional support. We then delve into the critical aspects of emotional speech in ECAs, examining: 1) the role of emotional speech, and 2) the techniques of concatenative synthesis and platform-based TTS generation in producing emotional speech. Additionally, we explore the concept of voice-appearance congruence, highlighting its significance in aligning an ECA's vocal characteristics with its visual appearance. Finally, we discuss key factors such as trust, perceived intelligence, and likability, which were integral to the measurements in our experiments.

2.1. Emotional Conversational Agents for Emotional Support

Embodied Conversational Agents (ECAs) have been extensively studied for their potential to provide emotional support, foster companionship, and alleviate loneliness [16] [31] [32]. These agents are particularly relevant in scenarios such as elderly care, where consistent and empathetic interactions can have a significant positive impact on user emotional well-being [33] [34] [35].

Pioneering work by Justine Cassell and her colleagues laid the foundation for this field by demonstrating the importance of integrating multimodal interaction techniques, such as speech, facial expressions, gestures, and other nonverbal cues, to create virtual agents capable of realistic and engaging human-like communication [17]. Building on these seminal insights, subsequent research has explored how ECAs can be designed to build and sustain long-term social and emotional connections with users, thushancing the overall user experience.

For example, Bickmore and Picard (2005) introduced relational agents that leverage strategies such as empathetic feedback, personalized dialogue, and adaptive social interaction to improve trust, participation, and adherence to health interventions [36] [37] [38] [39] [40]. Collectively, these studies underscore the promise of ECAs as versatile tools that can mitigate loneliness and enhance the emotional and social well-being of older adults, providing a solid foundation for further exploration and development in this rapidly evolving field.

In elderly care, ECAs are frequently used for daily activity support, cognitive stimulation, and companionship [41] [42]. Regular interactions with these agents have been shown to reduce loneliness and improve mood, especially in contexts where human interactions are limited. Their ability to simulate consistent, non-judgmental companionship highlights the importance of integrating emotional expressiveness, particularly through voice, into their design [43] [44] [45].

2.2. Emotional Speech in ECAs

2.2.1. The Role of Emotional Speech. The emotional tone of an ECA voice is a critical factor that influences user perception, emotional connection, and interaction results [46] [47] [48]. Emotional prosody, comprising rhythm, pitch, and stress, plays a significant role in the transmission of affective information [48] [49] [50]. Studies have shown that

Positive emotional tones (e.g., happy) enhance user satisfaction, perceived warmth, and engagement [51] [52]. Neutral tones are often interpreted as professional or calming, but may lack the emotional resonance required for empathetic support. This dichotomy between neutral and positive emotional tones forms the basis for evaluating how different emotional expressions impact the user experience, particularly among elderly users. In emotional support contexts, balancing emotional expressiveness is essential. Overly exaggerated emotional tones risk being perceived as insincere, while monotonic tones may appear unresponsive. Hence, carefully calibrated emotional speech is crucial for fostering trust and user comfort [53].

2.2.2. Concatenative Synthesis and Platform-Based TTS Generation in Emotional Speech. The ability to generate expressive emotional speech is heavily dependents on advancements in speech synthesis technologies [54]. In this study, we employ a concatenative synthesis approach that extends traditional methods by integrating scripted TTS generation with dynamic parameter adjustments. Specifically, predefined voice types are used in conjunction with modifications to pitch, rhythm, and prosody to modulate emotional expression. This hybrid approach offers several advantages. First, it produces high-quality, human-like speech; the combination of concatenative synthesis with manual parameter tuning yields smoother and more natural-sounding output, which is particularly well-suited for controlled utterances. Second, it enables enhanced emotional expression, as adjustments in acoustic parameters allow for fine-tuning of the speech to suit specific emotional contexts [55].

However, this approach also presents certain limitations. Platform-based parameter adjustments, while providing some flexibility, do not offer the same level of dynamic control over emotional tone as more advanced neural network-based methods [56]. Consequently, real-time adaptability and the ability to capture complex emotional shifts remain challenging. Moreover, reliance on predefined voice models and the limited control over fine-grained emotional nuances restrict the range of emotional expression and context adaptation.

Despite these limitations, concatenative synthesis combined with scripted TTS generation and manual parameter adjustment remains a widely used and accessible technique [57]. It is particularly suitable for controlled experimental studies that compare specific emotional tones, such as neutral versus happy expressions, rather than for exploring dynamic adaptability and complex emotional modulation. In contrast, advanced methods such as statistical parametric synthesis (SPSS) and end-to-end neural TTS models (e.g., Tacotron, WaveNet)—offer superior control over acoustic features, enabling real-time emotional modulation and greater contextual flexibility. These techniques address many of the inherent limitations of concatenative approaches [56].

2.3. The Role of Voice-Appearance Congruence

The congruence between an ECA's vocal characteristics and its visual appearance plays a pivotal role in shaping user perception and establishing trust. Research indicates that mismatched gender cues between an agent's voice and appearance can lead to cognitive dissonance, reducing engagement and undermining credibility [58]. Aligning these gender signals not only enhances the agent's anthropomorphic qualities but also creates a more seamless and natural interaction experience that fosters user comfort and trust [59]. Such consistency helps set clear expectations and reinforces the authenticity of the interaction, which is crucial for building a credible and relatable conversational agent.

Equally important is the emotional alignment across both visual and auditory modalities. An ECA that exhibits a happy facial expression paired with a neutral or mismatched vocal tone can appear inconsistent, thereby diminishing its emotional credibility. Conversely, when vocal tone and visual expressions are well synchronized, users are more likely to develop a strong emotional connection and feel that the agent is genuinely supportive [60] [61]. This multimodal congruence is essential for creating a compelling and emotionally intelligent conversational agent, one that delivers effective emotional support and fosters lasting engagement [62] [63]. These findings underscore the necessity of integrating coherent auditory and visual cues in ECA design to maximize both functional effectiveness and emotional impact.

2.4. Trust, Perceived Intelligence, and Likability

Trust and perceived intelligence are essential for user acceptance of embodied conversational agents (ECAs), especially in emotionally sensitive domains such as elderly care [64]. Trust is built on users' perceptions of the agent's competence, benevolence, and integrity (McKnight et al., 2011) [65], while perceived intelligence reflects its ability to respond appropriately within various contexts, thereby instilling confidence in its decision-making and problem-solving capabilities. Moreover, likability plays a critical role in shaping user engagement; ECAs that are viewed as friendly and agreeable can effectively reduce user anxiety, foster positive emotional responses, and encourage sustained interaction [66]. In this context, emotional speech is paramount, as it not only enhances the agent's warmth and empathy but also significantly boosts its responsiveness. By conveying nuanced emotional cues through speech, ECAs can forge deeper, more meaningful connections with users, ultimately enhancing their overall effectiveness in delivering emotional support and care [67].

3. The Design of the ECA

This section outlines the design and implementation of the Embodied Conversational Agent (ECA) used to explore the impact of the emotional tone of voice and the congruence of gender appearance on the experiences of elderly users. The ECA was created using advanced tools, including Unreal Engine's MetaHuman Creator for visual design, Live Link Face for real-time facial motion capture [68] [69], and Eleven Labs Text-to-Speech (TTS) for voice synthesis [70]. These technologies were integrated to produce high-quality, controlled video stimuli that systematically manipulated the experimental variables.

3.1. Visual Design of the ECA

The visual design of the ECA took advantage of Unreal Engine's MetaHuman Creator to produce realistic, human-like avatars with lifelike facial expressions and nuanced gestures [71]. A female avatar was used across all conditions to maintain consistency, with visual features adjusted for experimental requirements. For gender-appearance congruence, the avatar's visual gender (e.g., hairstyle and facial features) was matched or mismatched with the voice gender, resulting in four distinct conditions: matched (female voice + female avatar) and mismatched (male voice + female avatar). This manipulation allowed for an analysis of the interplay between visual and auditory gender cues.

Facial animations were created using the Live Link Face app, which captured real-time facial motion data synchronized with pre-recorded voice tracks. Subtle adjustments were made to ensure alignment with emotional tone; for instance, positive-tone ECAs exhibited slight smiles and dynamic eye movements, while neutral-tone ECAs maintained steady, composed expressions. To ensure natural interaction, the ECA videos were rendered against a neutral background with soft lighting, minimizing external distractions and highlighting the expressive qualities of ECA.

3.2. Voice Design and Emotional Tone Optimization

The ECA's voices were developed using Eleven Labs TTS, selected for its ability to produce expressive, high-quality audio. Neutral tones were characterized by a steady pace, minimal pitch variation [72], and calm intonation, while positive tones were enhanced with warmth and dynamic modulation [73]. Prosodic adjustments for positive voices included rhythmic variations, subtle rising intonations, and splicing techniques to emphasize key phrases and pauses [74] [75]. For example, phrases such as "It's SO NICE to see you today!" were refined to include stress markers and pauses, creating an engaging and conversational effect.

This approach leverages the strengths of concatenative synthesis to produce high-quality, human-like speech, while using manual parameter tuning to refine emotional expression. By combining predefined voice types with dynamic adjustments, we effectively modulated emotional tones to suit specific contexts. However, while this method works well in controlled settings, it has limitations in real-time adaptability and capturing complex emotional shifts—areas where advanced neural methods like WaveNet or Tacotron offer greater flexibility [56]. Despite these limitations, the combination of scripted TTS generation and manual adjustments proved effective for creating distinct neutral and positive tones, making it a practical choice for this study.

To validate the emotional quality and alignment of the synthesized voices, a pilot test was conducted with university students. Participants evaluated the naturalness, emotional tone, and congruence with visual features. Feedback informed iterative refinements, improving clarity and emotional resonance to ensure the voices were suitable for elderly users.

3.3. Video Production and Stimuli Creation

The final video stimuli were produced in Unreal Engine, combining the lifelike visual design of MetaHuman avatars with expressive motion capture and optimized voice tracks. Each video featured a pre-scripted 5-minute interaction focusing on neutral topics such as hobbies, daily routines, and general well-being, ensuring the content did not introduce external emotional influences. Positive-tone scripts incorporated rhythmic variations and pauses to enhance emotional engagement, while neutral-tone scripts maintained a steady flow to reflect composure.

The rendered videos underwent rigorous quality assurance to validate synchronization between voice and facial animations, emotional consistency, and alignment with experimental conditions. The output consisted of four high-quality MP4 videos tailored to the study's objectives, providing a controlled and immersive stimulus for elderly participants. By integrating cutting-edge TTS, motion capture, and rendering technologies, the ECA design ensured precision and realism, aligning with the study's experimental framework.

4. Methodology

4.1. Overview

This study investigates the impact of emotional tone of voice and gender-appearance congruence on elderly users' experiences with Embodied Conversational Agents (ECAs), aiming to address their emotional and social needs. Moving beyond previous research that predominantly relied on observational methodologies, this study emphasizes active user engagement to simulate real-world interaction dynamics and capture more nuanced user responses.

The study comprises two independent experiments: one focusing on emotional intonation and the other on gender-appearance consistency. In the first experiment, a two-group control design is employed to evaluate the impact of emotional tone (positive vs. neutral) on elderly users' emotional

responses, sense of companionship, feelings of loneliness, trust, perceived intelligence, and likability. Participants are randomly assigned to one of the two groups, interacting with the ECA using either a positive or neutral emotional tone. In the second experiment, the focus shifts to gender-appearance consistency, examining the effects of matching or mismatching the virtual agent's voice and appearance gender. Participants are divided into two groups: one with consistent gender presentation and the other with inconsistent gender presentation.

Both experiments employ standardized video-based stimuli, developed using advanced tools such as text-to-speech (TTS) and MetaHuman-powered visual avatars, ensuring consistency across conditions. The studies collect both quantitative and qualitative data, with participants completing various questionnaires to assess emotional responses and interaction experiences. The controlled approach allows for precise isolation of the experimental variables, enabling robust analysis of the main and interaction effects, and providing meaningful insights into how emotional tone and gender-appearance congruence influence elderly users' engagement with ECAs. To explore these questions, this study formulates the following hypotheses based on existing literature and the identified research gaps:

H1: ECAs with emotionally expressive tones will elicit stronger emotional reactions and greater perceived companionship compared to those with neutral tones. H2: Gender-congruent ECAs will be perceived as more trustworthy, intelligent, and likable than gender-incongruent ECAs. H3: The combination of emotionally expressive tones and gender-congruent appearance will produce the most positive outcomes across all user experience metrics.

4.2. Participants

The study was conducted by setting up experimental conditions on the Credamo platform, which facilitated data collection and analysis. Participants were recruited through the Prolific crowdsourcing platform, where a total of 86 elderly individuals aged 50-80 years were selected to ensure diversity in gender, socioeconomic status, educational background, and technological proficiency. Participants were compensated at a rate of €6 per hour to encourage engagement and completion of the study. The Credamo platform was used to design and administer the study tasks, including video viewing and questionnaire completion, while ensuring seamless integration with Prolific for participant recruitment and task fulfillment. To qualify, participants had to meet specific inclusion criteria, such as digital literacy (proficiency in using smartphones, tablets, or computers), cognitive health (absence of severe cognitive impairments or neurodegenerative conditions), and language proficiency (ability to understand the study materials). Ethical approval was obtained from the institutional review board, and informed consent was collected from all participants before data collection began. This integrated approach, leveraging Credamo for experimental design and data analysis alongside Prolific for recruitment, ensured a robust and efficient research process with a representative sample.

4.3. Experimental Design

This study comprises two independent experiments designed to examine the effects of emotional intonation and gender-appearance consistency on users' interactions with Embodied Conversational Agents (ECAs). Each experiment employs a two-group control design, focusing on a single independent variable to isolate its impact on the user experience.

In the first experiment, the focus is on emotional intonation, specifically comparing neutral versus positive emotional tones. The objective is to assess how these different intonations influence elderly users' emotional responses, sense of companionship, feelings of loneliness, trust, perceived intelligence, and likability towards the ECA. Participants are randomly assigned to one of two groups: one interacts with the ECA using a neutral emotional tone, while the other engages with the ECA using a positive emotional tone. This setup allows for the evaluation of the main effects of emotional intonation on various aspects of the user experience.

The second experiment investigates gender-appearance consistency by examining whether the alignment or mismatch between the ECA's voice gender and its virtual appearance affects user interactions. Participants are randomly divided into two groups: one group interacts with an ECA where the voice gender matches the virtual appearance (e.g., a female voice paired with a female virtual appearance), and the other group interacts with an ECA where the voice gender does not match the virtual appearance (e.g., a male voice paired with a female virtual appearance). This design aims to determine the impact of gender-appearance alignment on the naturalness and comfort of the interaction.

Both experiments utilize video-based stimuli to ensure control over external variables and maintain consistency across experimental conditions. This methodological approach facilitates a robust analysis by minimizing potential confounding factors, thereby enhancing the accuracy and reproducibility of the findings. By conducting these two separate experiments, the study effectively isolates and examines the individual effects of emotional intonation and gender-appearance consistency, providing a clearer understanding of how each factor independently influences elderly users' emotional and perceptual responses to ECAs.

4.4. Materials

The experiment included four conditions: a female ECA with a neutral voice, a male ECA with a neutral voice, a female ECA with a positive voice, and a male ECA with a positive voice. To ensure consistency, the same female MetaHuman avatar was used across all conditions, with only the vocal tone and emotional expressiveness systematically adjusted to reflect the experimental variables.

4.5. Procedure

Participants were briefed on the study's objectives and procedures. They were assured of the confidentiality and voluntary nature of participation. Written informed consent was obtained before beginning the study.

4.5.1. Pre-Experiment Validation and Refinements. Before deploying the surveys in the main experiments, a preliminary evaluation of voice samples was conducted with 10 university students. This pre-experiment aimed to assess the key attributes of emotional tone recognition, naturalness, and gender consistency in the voice samples. Results from an ANOVA revealed statistically significant differences in most evaluation dimensions, confirming the sensitivity of the survey items to variations in voice design. However, results for dimensions like "suitability for elderly users" and "emotional resonance" showed no significant differences between voice samples, suggesting areas for refinement. Based on these findings, adjustments were made to the voice parameters and the survey questions. For instance, the inclusion of pauses ("...") and emphasis markers (e.g., "SO MANY feelings!") in the voice scripts was refined to enhance emotional expression and resonance. Additionally, the surveys were updated to include more targeted qualitative questions about emotional and social impacts to ensure comprehensive feedback on the user experience.

4.5.2. Pre-Viewing Assessment. Baseline measures were collected prior to the viewing phase to control for individual differences. These measures included demographic data such as age, gender, education, and socioeconomic status. Additionally, participants' pre-existing emotional states were assessed using tools like the UCLA Loneliness Scale, which allowed for the measurement of loneliness levels before the interaction with the virtual agent. This pre-experiment data helped establish a baseline for comparison with post-viewing assessments.

4.5.3. Viewing Phase. Participants were randomly assigned to one of two experimental conditions based on the two separate experiments. In the first experiment, which focused on emotional tone of voice, participants were shown a 5-minute video in which the virtual agent interacted using either a positive or neutral emotional tone. In the second experiment, which examined gender-appearance congruence, participants watched an 5-minute video where the virtual agent's gender voice and appearance either matched or mismatched. The content of both videos remained neutral, covering topics such as daily life, hobbies, and past experiences to avoid introducing additional emotional variables. The goal was to ensure that the videos remained neutral while focusing solely on the variations in emotional tone or gender congruence.

4.5.4. Post-Viewing Assessment. After viewing the videos, participants completed the appropriate questionnaire based on their assigned experimental condition. Those in the emotional tone of voice experiment completed the Elderly Users' Virtual Agent Experience Questionnaire (EVAQ), which assessed emotional reactions, companionship, loneliness, trustworthiness, perceived intelligence, and likability. Those in the gender-appearance congruence experiment filled out the Virtual Gender-appearance Matching Questionnaire (VGMQ), evaluating the same constructs with an additional focus on how gender congruence influenced their perceptions. Participants rated each item on a 7-point Likert scale and provided qualitative feedback through open-ended questions, describing how the emotional tone or gender-appearance congruence of the virtual agent affected their experience.

4.5.5. Debriefing. Following the completion of the assessments, participants were thanked for their involvement in the study. They were provided with contact information for any follow-up inquiries and informed that they could reach out for any additional information related to the study. Moreover, resources for emotional support were offered in case participants felt the need for further assistance following the experiment. This step ensured that participants felt supported after their participation, and it concluded the study process in a professional and ethical manner.

4.6. Measurement Instruments

The evaluation of virtual agent interactions in this study was conducted using two custom-designed surveys: the Elderly Users' Virtual Agent Emotional Tone Experience Questionnaire (EVAQ) and the Elderly Users' Virtual Agent Gender Matching Experience Questionnaire (VGMQ). These instruments measure the impact of vocal characteristics and visual features of virtual agents on emotional well-being, social connectedness, and user experience. The design combines quantitative Likert-scale items and qualitative open-ended questions to ensure comprehensive data collection.

Both surveys are divided into six sections: emotional reactions, sense of companionship and loneliness, perceived usefulness, trustworthiness, perceived intelligence, and likability. Grounded in frameworks like the UCLA Loneliness Scale [76], Technology Acceptance Model [77], and Trust in Human-Robot Interaction [78], these surveys ensure reliability and contextual relevance.

The EVAQ examines the influence of emotional tone (neutral vs. positive) on user experience, emphasizing emotional reactions and companionship [79] [80]. For instance, participants rate statements such as "The virtual agent's vocal tone made me feel relaxed and at ease." The VGMQ assesses gender-appearance consistency, focusing on naturalness, comfort, and trustworthiness, with items like "The virtual agent's voice is consistent with its appearance gender."

Tailored to the experiments, these instruments provide precise evaluation of each variable while capturing holistic insights into elderly users' experiences with virtual agents.

Key frameworks ensure robust, validated analysis, informing the design of effective ECAs.

4.7. Data Analysis

The questionnaire data were analyzed using both descriptive and inferential statistical methods. Cronbach's alpha was calculated to assess the internal reliability of the Elderly Users' Virtual Agent Emotional Tone Experience Questionnaire (EVAQ) and the Virtual Gender-Appearance Matching Questionnaire (VGMQ). To examine the effects of emotional tone and gender-appearance congruence, one-way ANOVAs were conducted separately for each experiment. The emotional tone experiment compared participants' responses between the neutral vs. positive tone conditions, while the gender-appearance congruence experiment analyzed differences between the congruent vs. incongruent gender-appearance pairings. Effect sizes (partial eta squared) and Bonferroni corrections were applied to account for multiple comparisons. In addition, Pearson correlation analyses explored relationships between key perception factors, while linear regression models assessed whether voice tone or gender congruence significantly predicted companionship and loneliness reduction.

Qualitative data from open-ended responses were analyzed using thematic analysis to identify recurring patterns in participant experiences. Responses were categorized into key themes such as emotional engagement, naturalness of interaction, and trust in virtual agents. To ensure reliability, two independent researchers coded a subset of responses, and Cohen's kappa was calculated to measure inter-coder agreement. Differences in participant experiences across conditions were examined to understand how variations in voice tone and gender congruence influenced perceptions of the virtual agent.

A mixed-methods approach was used to integrate quantitative and qualitative findings, providing a comprehensive evaluation of user experiences. Statistical results were cross-checked with qualitative insights to identify converging patterns, while discrepancies were further explored for deeper interpretation. This combination of methods ensured a robust analysis of how emotional tone and gender-appearance congruence influenced user engagement, trust, and emotional connection with the virtual agent.

4.8. Ethical Considerations

Participants were informed about the nature of the study and the video-based presentation of ECAs. The experiment was designed to minimize any potential emotional discomfort, and participants were assured of their right to withdraw at any time. Anonymity and confidentiality of their responses were upheld throughout the study.

5. Results

5.1. Users' Virtual Agent Emotional Tone Experience

5.2. Users' Virtual Agent Gender Matching Experience

5.3. Summary

6. Discussion

Our study investigated the effects of voice emotional tone and voice–appearance gender matching on the user experience of elderly individuals interacting with conversational agents. This discussion is organized into three subsections: the impact of voice emotional tone, the effect of voice–appearance gender matching, and an integrated discussion with design implications, all supported by our experimental findings and user feedback.

6.1. Impact of Voice Emotional Tone on User Experience

In Experiment 1, we compared user experiences when interacting with a conversational agent that employed a neutral, natural female voice versus one with a positively modulated tone. The findings indicate that a neutral voice tone enhances users' emotional responses by making them feel happier, more relaxed, and more trusting during interactions. Participants noted that a natural tone creates a conversational experience that is more genuine and closely resembles real human interaction, thereby encouraging openness and emotional sharing. In contrast, while an overly positive tone aims to convey positive emotions, it was sometimes perceived as forced or exaggerated, which may detract from the authenticity of the interaction.

6.2. Impact of Voice–Appearance Gender Matching on User Experience

Experiment 2 focused on the importance of maintaining consistency between the agent's voice and visual appearance. The results clearly demonstrate that when the voice aligns with the agent's visual representation (i.e., a female voice paired with a female avatar), users report a higher sense of naturalness, comfort, and trust. This congruence enhances initial impressions and reinforces the agent's identity, leading to a more coherent and engaging interaction. However, while the matching condition significantly improved basic emotional responses, its influence on deeper dimensions such as companionship and perceived intelligence was less pronounced, suggesting that these more complex aspects may be moderated by additional factors.

6.3. Integrated Discussion and Design Implications

In summary, our findings underscore several key insights for designing effective voice-based conversational agents. A neutral, natural voice tone is essential for eliciting positive emotional responses and building trust, while maintaining consistency between voice and appearance is critical for establishing a coherent and comfortable user experience. Although these design choices clearly enhance basic affective measures, our results also indicate that deeper aspects such as companionship and perceived intelligence might benefit from further integration of additional modalities, including facial expressions, gesture animations, and context-sensitive adaptive responses. These insights collectively inform a holistic design framework that emphasizes naturalness, multimodal integration, and adaptive emotional delivery, paving the way for more engaging, emotionally intelligent conversational agents tailored to the needs of elderly users.

7. Limitation and Future Work

Although our findings provide robust evidence for the effects of voice tone and voice–appearance gender matching on user experience, several limitations warrant discussion and offer promising avenues for future research. First, the current study primarily relied on self-report questionnaires, which, while valuable, are subject to biases that could be mitigated by incorporating objective behavioral metrics and physiological measures. Second, our investigation focused solely on the isolated effects of voice tone and gender matching, yet real-world interactions with conversational agents are inherently multimodal. Future research should extend this work by examining the synergistic effects of additional modalities—such as facial expressions, gesture animations, and adaptive content in dynamic contexts—to enhance the realism and emotional depth of the agents.

In addition, the current experiment employed a TTS-based concatenative synthesis approach to generate emotional speech. While this method provides a useful baseline for exploring user responses, it is limited in delivering nuanced and dynamically adjustable emotional expressions. Future work should integrate advanced emotional speech synthesis techniques, such as end-to-end neural TTS models (e.g., Tacotron 2, FastSpeech) and state-of-the-art emotional speech conversion methods (e.g., CycleGAN, VAE-based models), to allow for real-time modulation of acoustic features like pitch, energy, and rhythm. Incorporating reinforcement learning-based adaptive mechanisms could further enable the system to personalize emotional delivery based on real-time user feedback, thereby enhancing naturalness and contextual relevance.

Looking ahead, design improvements can focus on several key directions. Enhancing the virtual agent's emotional expressiveness—through refined intonation, dynamic facial expressions, and context-sensitive adaptations—can bolster user companionship and positive affect. Strengthening the agent's intelligent response capabilities is essential to offer a more professional and tailored interaction experience. Moreover, optimizing the agent's ability to soothe users' emotions by conveying more positive energy through carefully designed speech and semantic content may further alleviate feelings of loneliness. Finally, reinforcing user trust and reliability by integrating professional information support and nuanced emotional feedback can enhance the overall credibility of the agent. Overall, while our current design demonstrates strong performance in terms of basic consistency and comfort, further exploration into deeper emotional interaction and enhanced intelligence is necessary to realize a more comprehensive and engaging user experience.

8. Conclusion

In conclusion, our study demonstrates that a neutral, natural voice tone significantly enhances users' positive emotional responses—including happiness, relaxation, and trust—compared to an overly positive tone, while maintaining consistency between voice and appearance (i.e., gender matching) is critical for establishing a coherent and comfortable interaction with conversational agents. Although our findings reveal clear improvements in basic affective measures, they also suggest that deeper dimensions such as companionship and perceived intelligence may benefit from the integration of additional modalities, such as facial expressions, gesture animations, and context-sensitive responses. Building on these insights, future virtual agent designs should focus on refining emotional expressiveness through advanced, adaptive speech synthesis and enhancing intelligent response capabilities to deliver more personalized, supportive, and trustworthy interactions, ultimately paving the way for more natural, engaging, and emotionally intelligent conversational agents.

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